

## A. Sleeping Through Classes

time limit per test: 1 second  
 memory limit per test: 256 megabytes

You have  $n$  classes today, which are numbered from 1 to  $n$ .

The classes are described by a binary string\*  $s$  of length  $n$ . We call class  $i$  *important* if and only if  $s_i = 1$ . For each *important* class, you **must** stay awake and listen to it.

You are very tired and wish to sleep through as many classes as possible. However, falling asleep takes time. If you listen to an *important* class  $i$ , then you cannot fall asleep for the next  $k$  classes, i.e., you must also stay awake in classes  $i + 1, i + 2, \dots, i + k$  (or until the end of the day, if fewer than  $k$  classes remain).

For classes that are not *important*, you may sleep through them unless the rule above forces you to stay awake.

Your task is to find out the maximum number of classes you can sleep through today.

\*A binary string is a string where each character is either 0 or 1.

### Input

Each test contains multiple test cases. The first line contains the number of test cases  $t$  ( $1 \leq t \leq 500$ ). The description of the test cases follows.

The first line of each test case contains two integers  $n$  and  $k$  ( $1 \leq n, k \leq 100$ ).

The second line of each test case contains the string  $s$  of length  $n$  ( $s_i = 0$  or  $1$ ).

### Output

For each test case, output a single integer — the maximum number of classes you can sleep through today.

### Example

| input    | Copy |
|----------|------|
| 4        |      |
| 4 1      |      |
| 1001     |      |
| 3 3      |      |
| 000      |      |
| 3 1      |      |
| 001      |      |
| 8 2      |      |
| 01000101 |      |
| output   | Copy |
| 1        |      |
| 3        |      |
| 2        |      |
| 2        |      |

### Note

In the first test case, you must listen to class 1 and class 4. After listening to class 1, you cannot fall asleep in class 2. So the only class you can sleep through is class 3.

In the second test case, you can sleep through all the classes.

### Codeforces Round 1068 (Div. 2)

Contest is running

01:45:48

Contestant



### → Submit?

Language: GNU G++17 7.3.0

Choose file:  No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts).  
 "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

### → Score table

|                           | Score |
|---------------------------|-------|
| <a href="#">Problem A</a> | 478   |
| <a href="#">Problem B</a> | 956   |
| <a href="#">Problem C</a> | 1195  |
| <a href="#">Problem D</a> | 1673  |
| <a href="#">Problem E</a> | 2151  |
| <a href="#">Problem F</a> | 2868  |
| Successful hack           | 100   |
| Unsuccessful hack         | -50   |
| Unsuccessful submission   | -50   |
| Resubmission              | -50   |

\* If you solve problem on 00:11 from the first attempt

In the fourth test case, you can only sleep through classes 1 and 5.

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